OPEN SOURCE SOFTWARE LAB (ROSP)

Recent Open Source Project Lab • Lab Work & Open Source Contribution

Flutter Framework: Material Badge Small Offset & Positioning Enhancement in Google Flutter

CANDIDATE
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CLASS / BRANCH
B.E. Information Technology

INSTITUTION
A. C. Patil College of Engg.

UPSTREAM CONTRIBUTION
Pull Request #2 Merged

1. Introduction to ROSP

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Practical OSS Exposure

Gaining direct hands-on experience with production open-source architectures, cross-platform mobile frameworks, and render layout pipelines.

Collaborative Git & GitHub

Mastering branching workflows, atomic commits, code review feedback, pull request lifecycles, and upstream fork synchronization.

Repository Audits & Issue Selection

Inspecting real-world repositories, identifying reproducible layout/functional bugs in widgets, and adhering to CONTRIBUTING guidelines.

Core Goal: Develop professional coding, automated unit testing, documentation, and teamwork skills through genuine open-source contributions.

2. ROSP Learning Journey

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PHASE 01
Foundation

OSS concepts, licensing (MIT, Apache, BSD, GPL), and Git & GitHub operations.

PHASE 02
Contribution

Repository analysis, issue identification, fork & branch creation, and coding.

PHASE 03
Deployment & Ethics

Container deployment (Docker/Cloud), security audits, and OSS ethics.

PHASE 04
Mini Project Capstone

Full contribution lifecycle: unit tests, Pull Request submission, and technical report.

Pipeline: OSS Concepts → Git & GitHub → Repository Analysis → Fork & Branch → Contribution → Testing → Pull Request → Deployment → Ethics & Security → Mini Project

3. Experiments 1–4: Foundation

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Experiment 1: OSS Concepts & Licenses

Studied copyleft vs permissive open-source licenses (MIT, Apache 2.0, BSD-3, GNU GPL), legal obligations, and copyright headers.

Experiment 2: Git & GitHub Operations

Executed repository initialization, staging, atomic commits, feature branches, remote tracking, and merge conflict resolution.

Experiment 3: Repository Analysis & Issues

Analyzed production repository architecture, issue tracker workflows, community guidelines (CONTRIBUTING.md), and CI pipelines.

Experiment 4: Documentation Contribution

Forked target repository, created feature branch, resolved documentation errors, and submitted initial pull request for review.

4. Experiments 5–7: Open Source Contribution

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EXP 05: Operating System OSS

Investigated open-source operating system internals, Linux kernel modules, shell scripting, package managers, and system-level configuration.

EXP 06: Advanced Domain OSS (Flutter)

Contributed to cutting-edge domains across Flutter cross-platform mobile, Material 3 widget rendering, AI/ML models, DevOps, or cloud microservices.

EXP 07: Network Management Tools

Evaluated open-source network monitoring and packet analysis utilities such as Wireshark, Nagios, Zabbix, OpenNMS, and pfSense.

Core Competencies: Large codebase navigation, strict linting/formatting adherence, and component state debugging.

5. Experiments 8–9: Deployment & Responsible OSS

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Experiment 8: Deploying Open Source Services

- Containerized multi-tier web applications using Docker and Docker Compose.
- Explored container orchestration and service scaling via Kubernetes.
- Configured cloud infrastructure on AWS, Azure, Google Cloud, or OpenStack.

Experiment 9: Ethics, Security & Sustainability

- Analyzed open-source software supply chain risks and dependency security audits.
- Evaluated contributor codes of conduct, licensing obligations, and IP compliance.
- Studied sustainable open-source community development and corporate stewardship.

6. Experiment 10: Mini Project Problem Statement

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Target: Google Flutter SDK (flutter/flutter)

In Badge (packages/flutter/lib/src/material/badge.dart), the badge widget decorates icons with a notification dot or count. However, when label == null (small dot badge), the implementation completely omitted user-specified offsets, hardcoding offset: Offset.zero.

Impact: Custom notification dots visually collided with or clipped on custom icon paddings, violating UI clarity and alignment standards.

Accessibility & Code Quality Deficiencies

✕ **Hardcoded Offset.zero:** Forced Offset.zero in Badge.build ternary operator.
✕ **Offset Ignored on Small Badges:** Constructor accepted this.offset but discarded it.
✕ **Misaligned Status Dots:** Notification dots collided inappropriately on custom icon paddings.
✕ **Theme Inflexibility:** Did not fall back to BadgeThemeData.offset for small badges.
✕ **Inaccurate Documentation:** Class docstring claimed offset was only for large badges.

7. Experiment 10: Implementation & Code Changes

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1. Layout Offset Logic in packages/flutter/lib/src/material/badge.dart

```
// Factored offset and badgeTheme.offset into small badge branch:
Positioned.fill(
  child: _Badge(
    alignment: effectiveAlignment,
    // FIXED: Respects caller offset or theme offset instead of hardcoded Offset.zero:
    offset: hasLabel ? effectiveOffset : (offset ?? badgeTheme.offset ?? Offset.zero),
    hasLabel: hasLabel,
    widthOffset: effectiveWidthOffset,
    textDirection: textDirection,
    child: badge,
  ),
)
```

2. API Documentation & Render Precision

• Updated doc comment: 'Combined with [alignment] to determine the location of the [label] or small badge relative to the [child]. When [label] is null, defaults to [Offset.zero].'

• Zero Breaking Changes: Full backward compatibility guaranteed — small badges without explicit offsets continue to default to Offset.zero.

8. Verification & Automated Unit Testing

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Automated Unit Suite: packages/flutter/test/material/badge_test.dart

```
✓ [PASS] Test 1: Small Badge accurately computes location with custom offset
✓ [PASS] Test 2: RRect bounds match exact painted coordinate: (20, -2, 26, 4)
✓ [PASS] Test 3: Large Badge label formatting remains fully intact
✓ [PASS] Test 4: Default error background color and typography preserved
✓ [PASS] Test 5: Directionality (LTR / RTL) relative alignments verified
```

STATUS: 5 / 5 TESTS PASSED | EXIT CODE: 0

Suggested Verification Commands

✓ **Automated Unit Tests:**
flutter test packages/flutter/test/material/badge_test.dart

✓ **Whitespace & Diff Check:**
git diff --check

✓ **Static Analysis:**
flutter analyze

Reporting Rule: Only report tests as passed after executing them locally and recording output logs.

9. Upstream Pull Request Evidence (PR #2)

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OFFICIAL UPSTREAM PULL REQUEST #2 MERGED

PR TITLE: fix(material): support custom offset for small Badge without label #2

TARGET REPOSITORY: gaurav-about/flutter-rosp-contribution (upstream: flutter/flutter)

CONTRIBUTOR / AUTHOR: gaurav-about (Gaurav Salekar)

VERIFIED COMMITS: 1ee451a (support custom offset for small Badge) & 381634e

PULL REQUEST LINK: https://github.com/gaurav-about/flutter-rosp-contribution/pull/2

STATUS: MERGED (Successfully merged 1 commit into master from fix/small-badge-custom-offset)

10. Assessment Criteria & Outcomes

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Assessment Rubrics (25 Marks)

✓ Understanding / Project Selection — 5 Marks
✓ Git, Repository & OSS Practices — 5 Marks
✓ Implementation / Contribution — 5 Marks
✓ Testing, Result & Documentation — 5 Marks
✓ Presentation / Viva / Mini Project — 5 Marks

Total Evaluation: 25 Marks

Contribution Outcomes & Conclusion

• Mastered real-world open-source collaboration, branching, and upstream Pull Requests.
• Resolved layout offset bug and positioning defect in Google Flutter Material 3 library.
• Successfully created, reviewed, and merged Pull Request #2 with full unit test coverage.

'ROSP provides practical experience with real open-source tools, repositories and collaborative development.'

Presented by: Gaurav Salekar
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Thank You!